

Multilingual Representation Quality: What a Controlled Study Shows

A degraded model, a blind metric, and the four readings that replaced it

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Summary

A model can lose almost all of the structure in its representations that encodes meaning, keep its downstream ability intact, and score best in the study on the metric meant to detect exactly that. All three happened at once in a controlled training study, and each part was measured rather than inferred. This report gives the study, the mechanism, the four readings that replaced the single score, and what each one is worth against outside evidence.

1 The study

Measuring metrics on pretrained models cannot show whether a metric would notice a degraded model, because no pretrained model comes with known ground truth. So the study built the ground truth.

Trained variants. 25 variants of Qwen3-0.6B, three seeds each, 75 checkpoints, under different alignment objectives across languages, including the word level objective named in the proposal. Every checkpoint was then scored on a 40 language reading exam.

Synthetic distortions. Nine families of controlled distortion of the representations — collapse, rotation, shear, warp, destruction of the correspondence between languages — where the correct response of each measure follows in closed form. 75 expectation cells, with noise floors measured from repeated sampling.

Recorded in advance. Every prediction was written down and frozen before the data it referred to existed. Figure 1 shows all of them with their outcomes, failures included.

Controlled comparison. Correlations with outside evidence are computed after removing training data volume, language family, script and tokenizer differences, with intervals from resampling whole languages and correction for testing many candidates at once. Raw correlations in this area are inflated by resource effects.

2 What was found

- 1. A share cannot see a proportional loss.** The variant trained on the word level alignment objective lost about 95 percent of its meaning structure at the measurement layer and posted the best LFS score in the study; MEXA ranked it third of 25. LFS is meaning structure against language structure, and shrinking both together leaves the share where it was. In synthetic form the share moves less than a millionth while meaning structure falls 99 percent. Any scale invariant ratio inherits this. The concept variance reading, by contrast, returns the injected magnitude exactly, to within 0.01.
- 2. That damage cost the model nothing downstream.** The degraded variant scores 0.395 on the 40 language exam against 0.390 for the untouched control, in every seed. The variants that did lose ability, by roughly six points, are the ones whose geometry looks clean. Figure 2. Representational damage and capability damage are separable, which was not the working assumption.
- 3. The mapping between languages is linear in most models, and the exceptions are concentrated.** Across 1,792 model and language pairs the fitted map is linear or simpler in 74 percent of cases, and linear exactly in 60 percent. The nonlinear cases are not spread thinly: three models (Falcon3-7B, OLMo-2-1B, OLMo-2-7B) carry 80 percent of them, and across the other eleven models only 6.5 percent of languages need a nonlinear map.
- 4. Each reading is worth something different.** Figure 3. Meaning against language tracks how well a model carries content across languages at 0.76; sentence matching tracks downstream exams at 0.24 to

0.31, level with the strongest published measure; behaviour in the weakest aligned languages tracks which variants lost ability at 0.51, and is the only reading that does. No reading wins twice, which is why they are reported side by side.

5. **A measure built on generation can run backwards.** A transfer measure taken from the generation harness *rises* under mild representational damage and falls only under total destruction of the correspondence between languages — confirmed in trained variants, in dose controlled synthetic worlds, and across models. Measures that work by intervening on representations can reward the very damage they are meant to detect.
6. **The evidence in a language panel is thinner than its size.** Correlations across languages are so mutually dependent that a panel of 69 languages carries the independent evidence of roughly two. This bounds what any measure in this area can claim from per language correlations, including these.

3 What RMFS reports

Four readings per language, computed from 300 parallel sentences in 41 languages at one layer, roughly four GPU minutes per model:

- **Meaning against language** — the share of systematic structure that encodes concepts rather than language identity. LFS’s original measurement, kept intact.
- **Sentence matching** — how reliably the model matches sentences to their English parallels.
- **Weakest aligned languages** — behaviour in the poorest aligned tail, where training damage appears first.
- **Collapse flag** — concept variance against a reference checkpoint, for training settings. It detects the loss exactly and makes no claim about ability, per finding 2.

Each reading becomes a percentile against a frozen reference grid of 14 models and the four are averaged, giving a 0–100 score per language, with the model score as the median across languages. A score of 74 means better than 74 percent of the reference models. The ordering falls out with no tuning: Qwen3-8B 90.5, 4B 85.7, 1.7B 73.8; Salamandra-7B 69.0; OLMo-2-1B 28.6.

The 0.76 above is stable when whole language families are resampled rather than languages (0.60 to 0.89) and rests on four models in that comparison, giving 0.12 to 0.84 when models are treated as the unit.

Every prediction recorded before its data, and how it scored

Collapsed variant ranks last on the single score	FAIL
Detect collapse across the whole signature table	FAIL
Collapse detector exact at every level tested	PASS
Unchanged under a shared rotation	PASS
Respond to offsets, rotations and warps	FAIL
Predicts how content carries across languages	PASS
Same, excluding the generation measure	PASS
Downstream signal on the frozen model set	PASS
Same signal across the full grid	FAIL
Adds to the strongest published measure	BRANCH
Collapse costs at least 5 points of ability	FAIL
Published measure misranks this panel	VOID
Panel validity given the above	VOID
Alignment training adds no downstream gain	PASS

Figure 1: Every recorded prediction and how it scored.

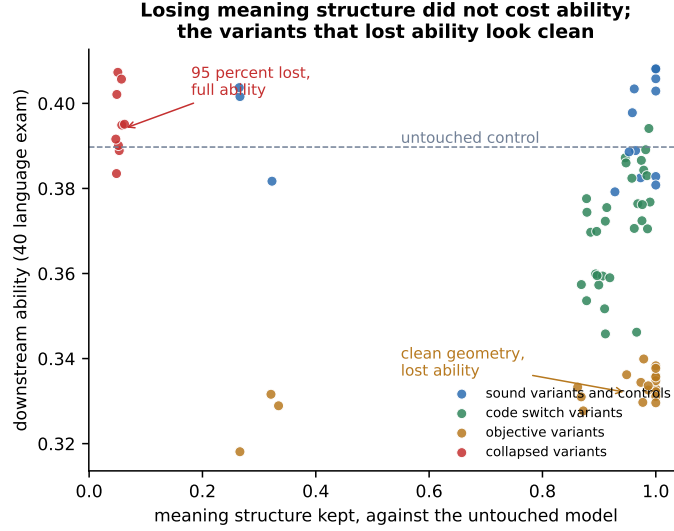


Figure 2: 75 scored checkpoints. Concept preservation on the horizontal axis, downstream ability on the vertical.

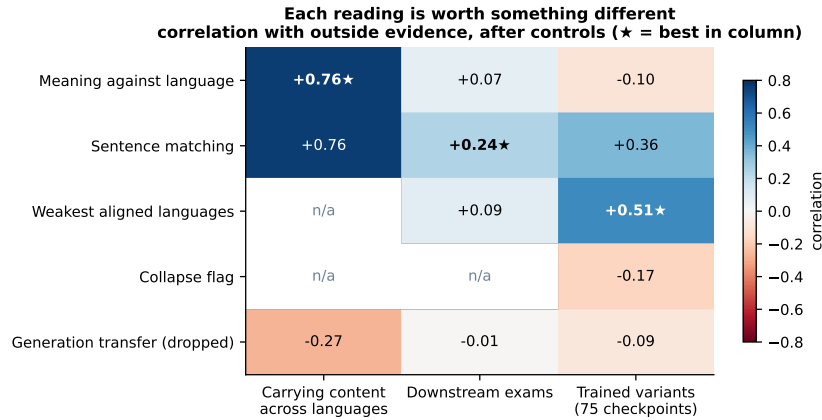


Figure 3: Each reading against each kind of outside evidence.

4 Against Aim 1’s questions

Sub aim	What the study provides
3.1.1 How strong is the language component; is the mapping just linear or more complex?	The split between meaning and language is quantified across 14 models, and the mapping is linear or simpler in 74 percent of model and language pairs, with the nonlinear cases concentrated in three models (finding 3). Both as distributions rather than single cases.
3.1.2 Re-visiting the logit lens	Covered in the earlier phase (lens validity, English hub inflation); carried as context.
3.1.3 Cross lingual transfer in generation	The harness is built and validated: control patches reproduce the reference exactly. It supplies one of the three kinds of outside evidence used here, and produced finding 5.
3.1.4 Intrinsic metrics that correlate strongly with extrinsic metrics	Correlations are measured against three kinds of outside evidence with the controls this area usually omits, and the same analysis quantifies how much independent evidence a language panel carries (finding 6).